Trainings

Preparatory Steps

⚠ WARNING ⚠

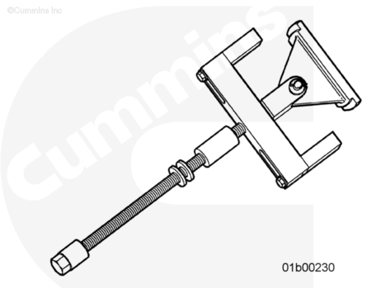
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html)
- Remove the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html)

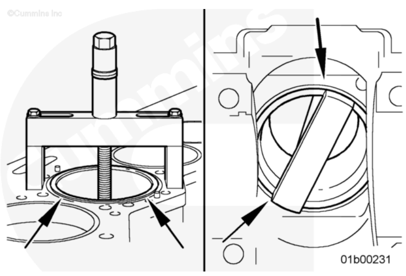
Remove

Use the cylinder liner puller, Part Number 3163745, and puller plate, Part Number 3162886, to remove the cylinder liner.

Wind down the threaded rod to extend the arm. Turn the arm to the side to enable passage through the liner.

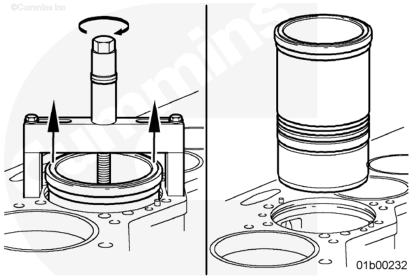


Install the cylinder liner puller, Part Number 3163745, in the cylinder liner. The puller feet **must not** touch the top of the liner. The puller arms **must** be positioned firmly on the bottom of the liner.



Turn the puller screw until the liner loosens in the block.

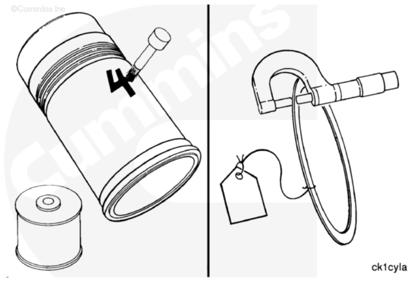
Remove the tool and the liner.



Use a liquid metal marker to mark the cylinder number and bank on each liner. Mark the cylinder liner on the camshaft side of the cylinder liner.

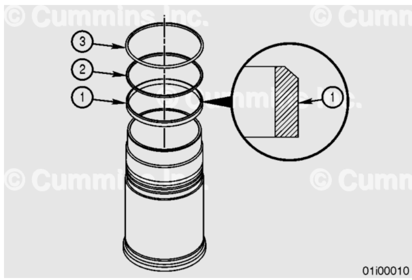
If sealing rings were used, use a tag to mark the cylinder number.

Measure in several places and record the thickness of the sealing rings used in each cylinder. The thickness of the sealing ring is one factor in determining liner protrusion. This information **must** be known when the liners are installed in the engine.



Remove and discard the two D-rings (1 and 2).

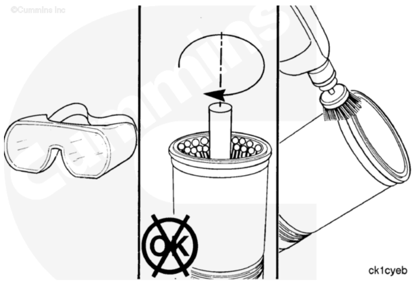
Remove and discard the crevice seal (3).



Clean and Inspect for Reuse

⚠ WARNING ⚠

To reduce the possibility of personal injury wear appropriate eye and face protection. Make sure the wire brush is rated for the RPM being used if the brush is motor driven.



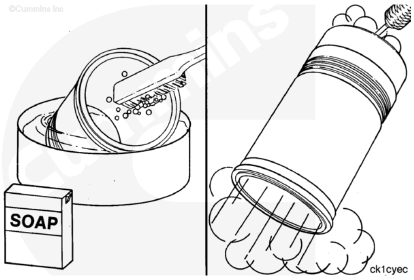
⚠ CAUTION ⚠

Do not use a hone, deglazing, or prebrushing to clean the cylinder liners. Abrasives can damage the finish and the crosshatch pattern and can contaminate the liner.

Use a high quality steel wire brush to clean the liner flange seating area.

⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

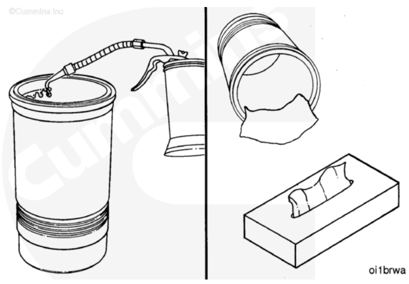
Use a nonmetallic bristle brush, detergent soap, and warm water to clean the inside of the liner.

Use a steam cleaner or solvent tank to clean the liners.

Dry with compressed air.

Lubricate the inside diameter of the cylinder liner with clean engine oil.

Allow the liner to soak for five to 10 minutes.

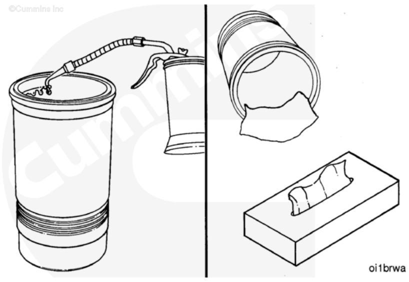


⚠ CAUTION ⚠

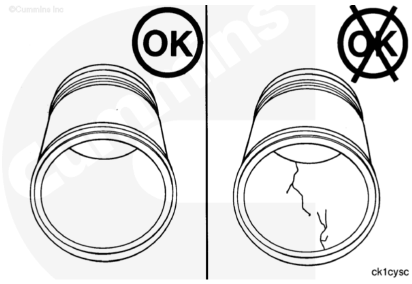
Do not use a cloth towel. Lint will cause severe engine damage.

Clean the inside diameter of the cylinder liner with a paper towel.

Repeat the cleaning steps until the paper towel does **not** show gray or black residue.

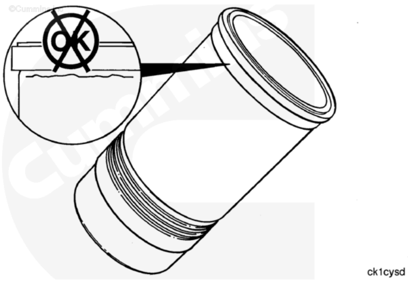


Inspect the liners for cracks on the inside and the outside diameters.



Inspect for cracks under the flange.

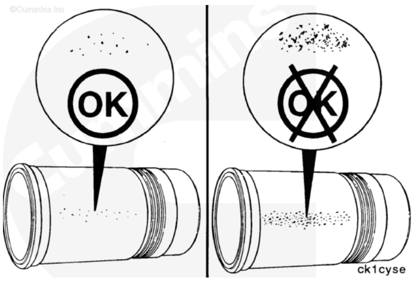
Note : Cracks can also be detected by using either magnetic inspection or the dye method.



Inspect the outside diameter for excessive corrosion or pitting.

Pits **must not** be more than 1.6 mm [0.063 in] deep.

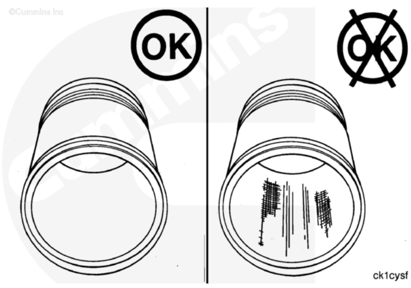
Replace the liner if the pits are too deep or if the corrosion can **not** be removed with an abrasive pad.



Inspect the inside diameters for vertical scratches deep enough to be felt with a fingernail.

If a fingernail catches in the scratch, the liner **must** be replaced.

Inspect the inside diameter for scuffing or scoring.

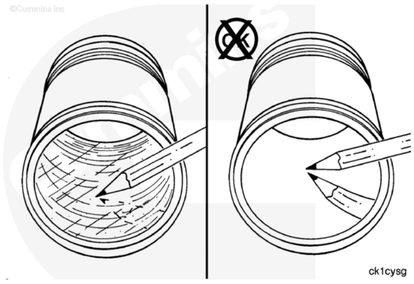


Inspect the inside diameter for liner bore polishing.

A moderate polish produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

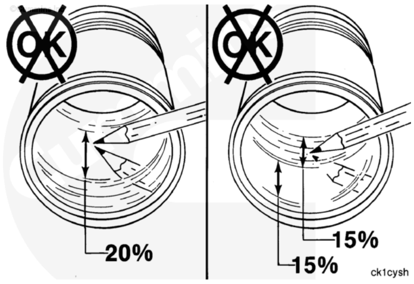
A heavy polish produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.

Reference Parts Reuse Guidelines, Bulletin Number 3810303, for further information on liner bore polishing.



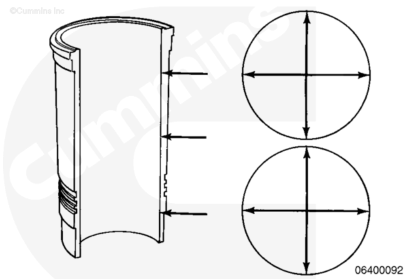
Replace the liner:

- If a heavy polish is present over 20 percent of the piston ring travel area
- If 30 percent of the piston ring travel area has both moderate and heavy polish, with one-half of the 30 percent (15 percent) consisting of heavy polish.



Note : The inside diameter of a new cylinder liner can be 0.015 mm [0.0006 inch] smaller than specifications because of the Lubrite™ coating.

Use a dial bore gauge to measure the inside diameter of the liner at the top, the bottom, and the middle of the piston ring travel area. Perform two measurements at each location. The measurements **must** be 90 degrees apart.

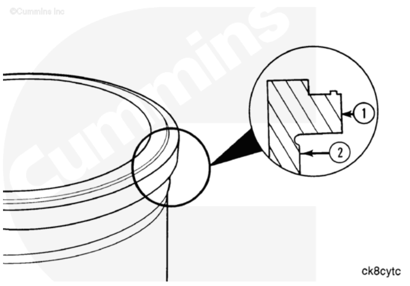


Cylinder Liner Inside Diameter		
mm		in
158.75	MIN	6.2495
158.78	MAX	6.2550

The cylinder liners are:

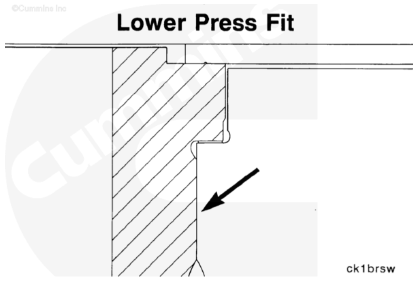
1. Flange outside diameter referred to as the upper press fit outside diameter.
2. Liner outside diameter referred to as lower press fit outside diameter.

For example, 20/20 OS means the upper press fit outside diameter (1) is 0.020 inch larger than the standard liner and the lower press fit outside diameter is 0.020 inch larger than the standard liner.

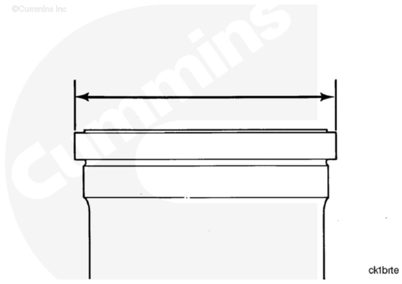


The liner design incorporates a press fit between the upper liner bore and the area of the liner directly below the liner flange. This is referred to as the lower press fit design.

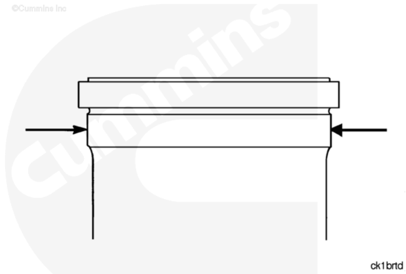
Cylinder liners with standard and oversize press fit diameters are available.



Measure the liner flange outside diameter.



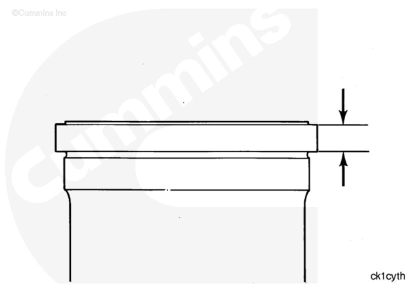
Measure the lower press fit diameter. Reference the following tables for liner sizes.



Mechanically Actuated Injectors							
		Upper Press Fit Liner Flange Diameter			Lower Press Fit Area Outside Diameter		
Part Number	Size (in)	mm		in	mm		in
4333984	-0.01/0	187.965	MIN	7.400197	180.165	MIN	7.10311
		188.015	MAX	7.402165	180.215	MAX	7.095079
4308809	Standard	188.189	MIN	7.409016	180.165	MIN	7.09311
		188.239	MAX	7.410984	180.215	MAX	7.095079
4371641	Oversize +.010/.010	188.443	MIN	7.419016	180.419	MIN	7.10311
		188.493	MAX	7.420984	180.469	MAX	7.105079

Mechanically Actuated Injectors							
		Upper Press Fit Liner Flange Diameter			Lower Press Fit Area Outside Diameter		
Part Number	Size (in)	mm		in	mm		in
4333987	Oversize +.020/.020	188.697	MIN	7.429016	180.671	MIN	7.113031
		188.747	MAX	7.430984	180.721	MAX	7.115
4371767	Oversize +.030/.030	188.951	MIN	7.439016	180.927	MIN	7.12311
		189.001	MAX	7.449084	180.977	MAX	7.125079
4371642	Oversize +.040/.040	189.205	MIN	7.449016	181.181	MIN	7.13311
		189.255	MAX	7.450984	181.231	MAX	7.135079
4333989	Oversize +.060/0	180.713	MIN	7.469016	180.165	MIN	7.09311
		180.763	MAX	7.470984	180.215	MAX	7.095079
4333988	Oversize +.060/.020	189.713	MIN	7.479016	180.671	MIN	7.113031
		189.763	MAX	7.470984	180.721	MAX	7.115

Measure the liner flange thickness.

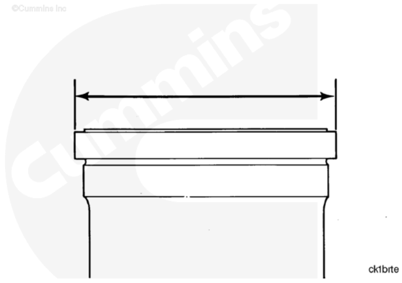


Mechanically Actuated Injectors Cylinder Liner Flange Thickness				
	mm		in	
Undersized Liner	7.747	MIN	0.305	
		7.773	MAX	0.306024
Standard and	13.398	MIN	0.52748	
Oversized Liner		13.424	MAX	0.528504

Measure the cylinder liner flange outside diameter.

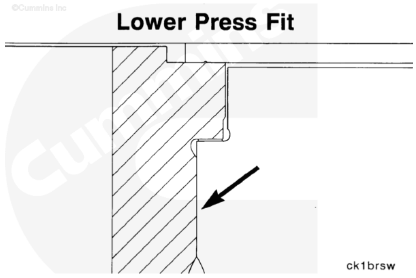
Upper Press Fit Cylinder Liner Flange Diameter			
	mm		in
Standard	188.19	MIN	7.409
	188.24	MAX	7.411
Oversize 20/20	188.70	MIN	7.429
	188.75	MAX	7.431

If the cylinder liner is **not** within specifications, the cylinder liner **must** be replaced.



The liner design incorporates a press fit between the upper liner bore and the area of the liner directly below the liner flange. This is referred to as the lower press fit design.

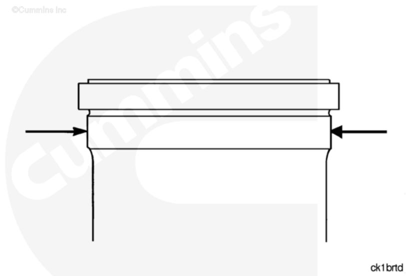
Cylinder liners with standard and oversize press fit diameters are available.



Measure the cylinder liner lower press fit diameter.

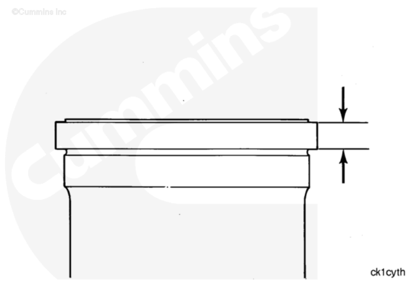
Cylinder Liner Lower Press Fit Area Outside Diameter			
	mm		in
Standard	180.16	MIN	7.093
	180.21	MAX	7.095
Oversize 20/20	180.67	MIN	7.113
	180.72	MAX	7.115

If the cylinder liner is **not** within specifications, the cylinder liner **must** be replaced.



Measure the cylinder liner flange thickness.

Cylinder Liner Flange Thickness		
mm		in
13.411	MIN	0.528
13.437	MAX	0.529

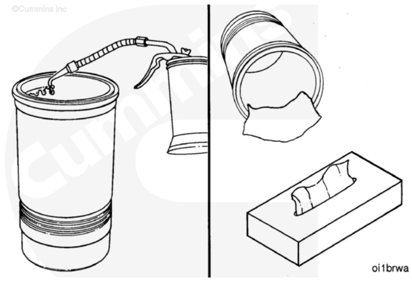


If the cylinder liner is **not** within specifications, the cylinder liner **must** be replaced.

Apply a thick film of clean 15W-40 oil to the bores of the liners for final cleaning. Leave the oil on for 5 to 10 minutes.

Use a clean, lint-free paper towel to wipe the oil from the bores until the black and gray deposits are removed.

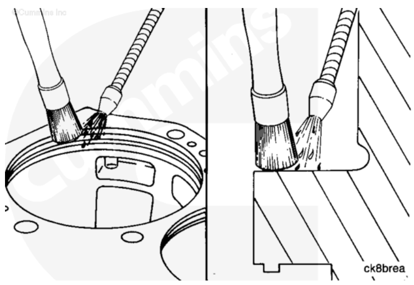
Do **not** place the liners in an area where dirty air flow can contaminate the liners.



Install

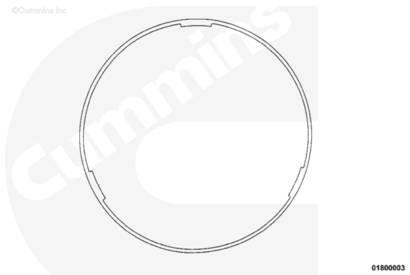
⚠ WARNING ⚠

When using solvents, acids, or alkaline material for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



Clean the bottom of the cylinder block cylinder liner flange with safety solvent.

The seal rings have three locating tabs on the inside diameter. The tabs have an interference fit to the liner lower press fit diameter to hold the seal ring in place during liner installation.



Install the seal rings.

The seal ring **must** be straight on the liner upon installation. Use finger pressure to press the seal ring near the tabs to fit the seal ring down and over the lower press fit diameter during installation.

This practice during installation of the seal ring will prevent deformation that will result in the seal ring **not** fitting squarely on the bottom of the liner flange.

Note : Some o-rings have a "D" shape cross section. This type of o-ring must be installed with the flat side against the cylinder liner.

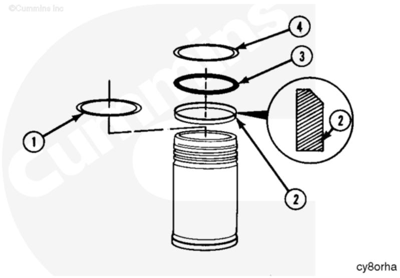
Install the liner, counter bore sealing ring (1).

If an upper crevice seal was used, install the upper crevice seal with the white side out.

Install the crevice seal. The beveled edge of the crevice seal (2) **must** be positioned as shown.

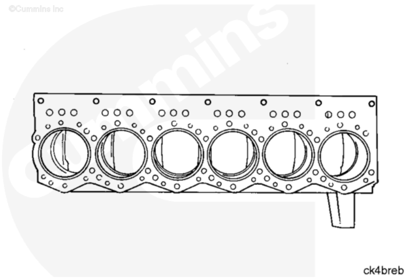
Install o-rings in the position shown. Use the mold mark on the o-ring to check if the o-ring is twisted.

- (3) Black o-ring
- (4) Red o-ring.



Use vegetable oil to lubricate the inside diameter of the packing ring bores.

Use hand pressure to press the cylinder liners into the block.



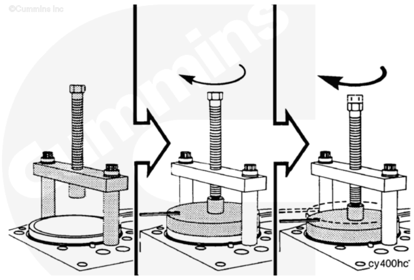
Use liner installation tool, Part Number 3375422, or equivalent, to install the bridge assembly and two cylinder head capscrews. Tighten the capscrews.

Torque Value: 45 n•m [34 ft-lb]

Install the pusher plate in the liner. Make sure it is aligned correctly in the liner.

Turn the pusher screw until it touches the plate. Turn the pusher screw until the liner flange touches the counterbore ledge.

Do **not** use more than 65 N•m [50 ft-lb] of torque. Remove the tool.

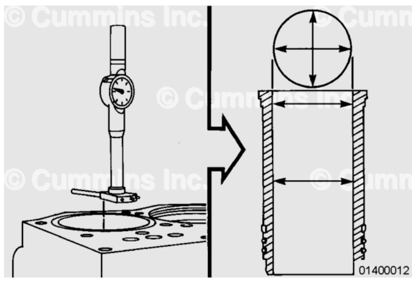


Note : New cylinder liners can be 0.005 to 0.015 mm [0.0002 to 0.0006 in] smaller than the minimum specifications because of the Lubrite coating.

Use a dial bore gauge and measure the inside diameter of the liner at the top, bottom, and middle of the liner.

Perform two measurements at each location. The measurements **must** be 90 degrees apart.

New Cylinder Liner Inside Diameter		
mm		in
158.750	MIN	6.250



New Cylinder Liner Inside Diameter		
mm		in
158.775	MAX	6.251

The inside diameter **must not** be more than 0.076 mm [0.003 in] out-of-round at the top two measurements.

If the inside diameter is more than 0.05 mm [0.002 in] out-of-round in the bottom measurement location, the liner **must** be removed. Check for a twisted o-ring.

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Measure the cylinder liner protrusion. Refer to Procedure 001-064 in Section 1. (/qs3/pubsys2/xml/en/procedures/18/18-001-064-tr.html)
- Install the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html)

Last Modified: 02-Jan-2019